PREFACE

In the realm of modern advanced driver assistance systems (ADAS) and autonomous driving, the precise and efficient detection of traffic signs is crucial. As the drive towards safer and more reliable self-driving vehicles intensifies, the need for robust algorithms capable of performing well under diverse real-world conditions—such as varying lighting, occlusion, weather, and perspectives—becomes increasingly important.

In 2011, the GTSRB showcased the ability of state-of-the-art algorithms to handle traffic sign classification effectively. Building on this success, the GTSDB was established to focus specifically on detecting the exact location of traffic signs in natural scenes. This benchmark underscores the necessity for robust detection methods to manage real-world environments' complexities.

Given the industrial significance and ongoing challenges in traffic sign detection, this project investigates the application of different iterations of the YOLO (htt34)algorithm for this purpose. YOLO is renowned for its speed and accuracy, making it an excellent candidate for real-time applications like ADAS.

The "Traffic Sign Detection using Different Versions of YOLO" project aims to compare the performance of four YOLO model versions: YOLOv3, YOLOv5, YOLOv7, and YOLOv8. Each version offers distinct enhancements and optimizations that could influence their efficacy in detecting various traffic sign categories, including prohibitory, danger, mandatory, and others.

By systematically training and assessing these models on a comprehensive dataset of traffic signs, the project seeks to identify which version strikes the best balance between precision and recall. The findings indicate that YOLOv8, developed by Ultralytics, outperforms the other models, achieving the highest mAP50-95 scores across all categories. To gauge their practical effectiveness, the evaluation considers precision, recall, and mean average precision (mAP) metrics.

Among the models evaluated, YOLOv8 emerged as the top performer, demonstrating superior precision and recall rates, with a mAP@50 of 0.984 and a mAP@50-95 of 0.843. This outstanding performance is attributed to YOLOv8's advanced architecture, which features improved feature extraction, better handling of small objects, and enhanced training protocols. These improvements enable YOLOv8 to detect traffic signs more accurately and consistently under various challenging conditions, including different lighting, occlusions, and other scenarios, making it the most effective model for traffic sign detection in this study.